Campbell Lab
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SOP #15
Western Blotting

Running the SDS-PAGE:

1. Prepare samples for loading and standard curve - SOP #14
2. Prepare 440 mL (**per gel**) of Bolt 1X running buffer according to Bolt protocol quick sheet.
3. Set up rig with metal prods from the electrophoresis clamps facing towards the middle (check that the lid fits). Pour running buffer into rig until just above (1 cm) the metal conducting plate.
4. Using gloves, open commercial gel package; rinse wells and discard comb/sticker. Don't touch anything else, as un-polymerized acrylamide is a **neurotoxin**. Place gel in rig, remove gloves and snap shut the clamp.
5. Discard gloves in special waste container for acrylamide gloves.
6. Load the samples (heated, spun and on ice) into the gel as well as a ladder lane. Ladder is a mixture of 0.5 uL MagicMark and 4.5 uL NovexSharp (both from Life Tech). Asymmetrical ladder loading is a good practice, it will help orient your blot during imaging.
7. Run the gels for 22-35 minutes at 200 V in 1X MOPS (proteins >70 kDA) or 1X MES buffer (everything else). Timing can be optimized thanks to visible NovexSharp pre-stained ladder; run the gel long enough until the size range for your protein of interest is halfway down the gel.

Meanwhile, prepare the items for transfer:
~200 mL of 1X transfer buffer (**per blot**); according to the Bolt protocol quick sheet.
a sheet of pre-cut PVDF with notched corner (or in rare cases nitrocellulose membrane)
a small volume (50 mL) of methanol for rehydration of PVDF (reusable)
2 sponges per blot
2 pre-cut filter papers per blot
a blot module cassette per blot
blot tweezers
a roller
a metal tool for plastic cracking
one small plastic dish (old pipette holder lid) and one medium size plastic dish (sandwich container)

Transfer Step:

1. Set up cookie sheet with instruments on it, the methanol in the smaller plastic dish & transfer buffer in the medium sized plastic container. Pour a bit of 1X transfer buffer onto the negatively charged module piece - this is where your transfer sandwich will go.
2. With gloves on, soak and massage the first sponge. Place it in the pre-wetted module piece. Be sure to expel all air bubbles - this is crucial. The roller may help. If transferring two gels, wet the second module and sponge the same way. Sponges go: longer edge along the bottom.
3. Soak and massage the top sponges now, then leave them in the buffer for use later.
4. Activate the PVDF in methanol for at least 30 seconds. Place the activated PVDF in the transfer bucket, making sure it **does not dry out**! It floats, so put the filter paper on top to keep it down.The methanol activation step is unnecessary when using nitrocellulose membrane.
5. Now you are ready to crack open the gels. Once your gloves touch polyacrylamide, they are considered contaminated and you need to limit what you touch, then discard them properly.
6. WEAR GLASSES for this step - neurotoxic plastic can go flying.
7. Take the metal tool and jam it between the two plates of the pre-cast gel, which should be facing up (shorter plate upwards) Twist to open. Remove shorter plate, set aside.
8. Cut off wells and pile them on your discarded plate. The gel matrix is sticky, so keep it sequestered as best as possible.

9. Place your first filter paper along the well line.
10. Flip the gel, and push the gel off the plate using the metal tool through the foot/location of sticker.
11. Place it on the cookie sheet, and cut off the foot. Lane #1 is on the far right.
12. Gently place the PVDF on the gel, notched side in upper right corner. Remove air bubbles.
13. Finish the sandwich with a second piece of filter paper.
14. Gently roll the sandwich with the plastic roller.
15. Using the metal tool, lift the sandwich and place it in the same orientation into the transfer module (on the sponge). Add the second sponge.
16. Close the module, place it in your new rig, feet facing you. Do the second cassette now if needed.
17. Fill the inner chamber of the cassette(s) with 1X transfer buffer (1 cm above electrode), and the space outside the module(s) with cold tap water. Fit the lid.
18. Transfer 60 minutes at 30V. You can optimize the voltage to 20V if you think the proteins are transferred right through the PVDF (this outcome depends on size, ask someone for your first blot)

Once finished, the blots can be blocked overnight in the fridge (or one hour at room temperature) in a 5% fat-free milk powder solution or 2% ECL blocking solution. You should prepare this solution while the transfer is happening. Each blot box holds 10 mL (20 mL for overnight incubation to account for evaporation). The blocking agent should be dissolved in 1X TBS-T (0.1%) **NOT WATER**.

Label all the boxes carefully. Once the transfer ends, carefully open the modules. The PVDF goes directly into the blocking solution **protein side up**. This is the side that was touching the gel. Don’t let the PVDF dry out.

The actual gel goes in the acrylamide waste bucket. Used filter paper can go in the green garbage, sponges should get rinsed multiple times in dH_2O then pressed to expel water (don’t bend or crush them). Rigs get rinsed in dH_2O.